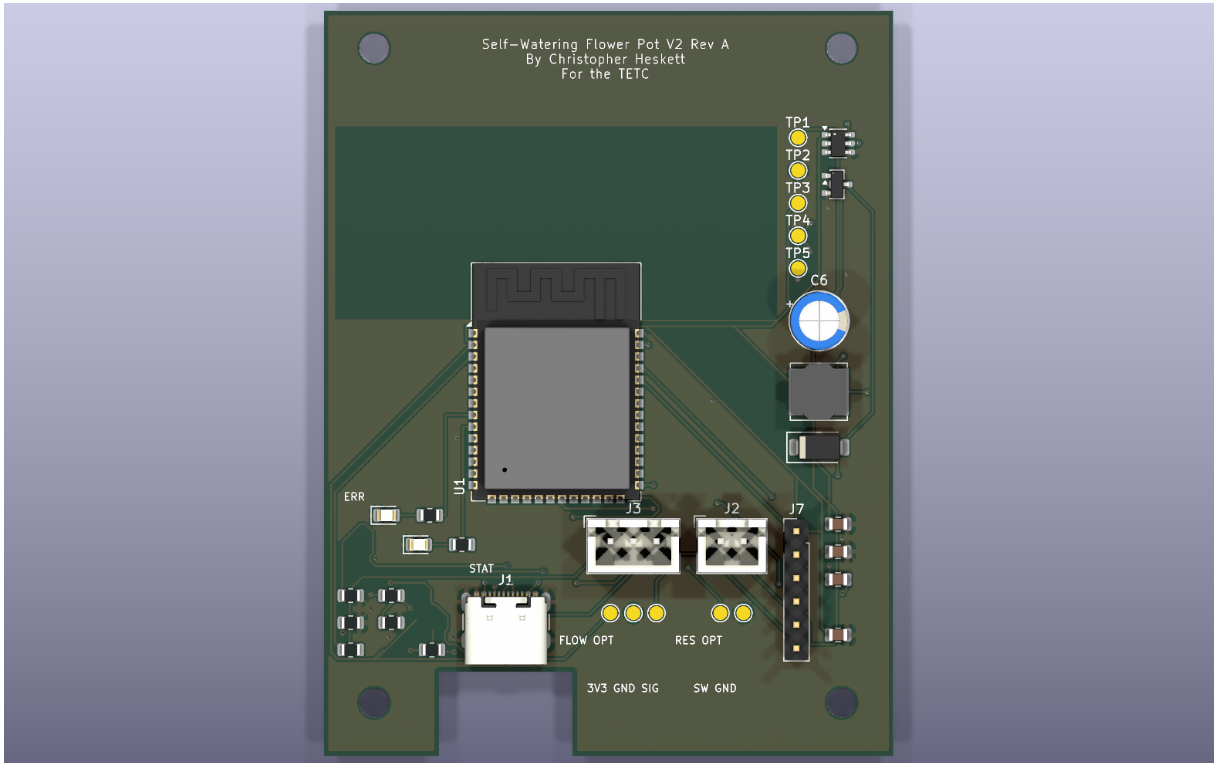
Self-Watering Flower Pot V2 - Board A Cost-Down Assembly Reference

Standalone printable placement and testpoint guide. This is intentionally separate from the BOM.



Build scope
Board A cost-down population only. Board B UI is deferred. Reservoir and flow features are exposed as optional pads, not required JST connectors.
Do not populate for minimum build
C4 optional ADC filter; older UI/reservoir/flow JST parts J4/J5/J6 and R6/R7/R8 are not part of this cost-down assembly.
First power checks
Before pump testing, measure TP1 to TP2 for +5 V and TP3 to TP2 for +3.3 V. Confirm TP4/PUMP_GATE stays low during reset/boot.

Suggested Assembly Order

Step	Install/check	Refs
1	USB and 3.3 V power section; verify rails before adding pump load.	J1, R1, R2, U2, L1, C1, C2, C3, C5, C6, TP1, TP2, TP3
2	MCU and fallback programming header.	U1, R3, J7
3	Pump driver, with pump still disconnected for first gate tests.	Q1, D1, R4, R5, J2, TP4
4	Moisture input and local status indicators.	J3, R10, TP5, D2, D3, R11, R12
5	Optional pad wiring only if firmware/settings enable it.	TP6, TP7, TP8, TP9, TP10

Component Placement: What Goes Where

All fitted parts are on the top side. Region names match the top-view image on page 1.

Ref	Part/value to place	Board area	Use / orientation note
C1	10uF 10V X7R input	bottom-right	Buck input capacitor; near U2/J1 power input.
C2	22uF 6.3V X7R output	bottom-right	Buck 3.3 V output capacitor.
C3	100nF 10V BST	bottom-right	AP63203 bootstrap capacitor.
C5	22uF 6.3V X7R output	bottom-right	Second buck 3.3 V output capacitor.
C6	100uF 10V pump bulk	middle-right	Pump/5 V bulk electrolytic; observe polarity.
D1	SS34 flyback	middle-right	Pump flyback diode; use SMA B340A/SS34-class part; observe stripe orientation.
D2	ERROR red	bottom-left	Error LED, red.
D3	STATUS green	bottom-left	Status LED, green.
J1	USB-C 5V + USB2	bottom-left	USB-C 5 V input/native USB connector.
J2	PUMP 5V	bottom-right	Pump JST-XH: pin 1 +5V/PUMP+, pin 2 PUMP_LOW/PUMP-.
J3	MOISTURE	bottom-center	Moisture JST-XH: pin 1 +3V3, pin 2 GND, pin 3 MOISTURE_ADC.
J7	DEBUG 3V3/GND/TX/RX/EN/BOOT	bottom-right	Fallback debug/program header: 3V3, GND, TXD0, RXD0, EN, IO0/BOOT.

Ref	Part/value to place	Board area	Use / orientation note
L1	3.9uH shielded >=3A Isat	middle-right	3.9 uH buck inductor.
Q1	AO3400A	top-right	AO3400A low-side pump MOSFET.
R1	5.1k CC1	bottom-left	USB-C CC1 5.1k sink resistor.
R2	5.1k CC2	bottom-left	USB-C CC2 5.1k sink resistor.
R3	10k EN pullup	bottom-left	ESP32 EN 10k pullup.
R4	220R gate	bottom-left	Pump gate series resistor.
R5	100k pulldown	bottom-left	Pump gate pulldown; keeps pump off at reset.
R10	1k ADC series	bottom-left	Moisture ADC series resistor.
R11	330R error LED	bottom-left	Error LED current-limit resistor.
R12	330R status LED	bottom-left	Status LED current-limit resistor.
U1	ESP32-S3-WROOM-1-N8	middle-center	ESP32-S3-WROOM-1-N8 module; antenna at top, keepout clear.
U2	AP63203WU-7	top-right	AP63203WU-7 3.3 V buck regulator.

Connector Pinouts

Connector	Pins / meaning	Assembly note
J1 USB-C	VBUS=+5V, GND=GND, CC1/CC2 use R1/R2 5.1k; USB D+/D- for native ESP32 USB.	Use a real USB-C data cable for flashing.
J2 Pump	Pin 1 = +5V/PUMP+, Pin 2 = PUMP_LOW/PUMP- switched by Q1.	Do not connect pump until TP1/TP3 and TP4 behavior are verified.
J3 Moisture	Pin 1 = +3V3, Pin 2 = GND, Pin 3 = MOISTURE_ADC through R10.	Sensor output must stay at or below 3.3 V.
J7 Debug	Pin 1 = +3V3, 2 = GND, 3 = TXD0, 4 = RXD0, 5 = EN, 6 = IO0/BOOT.	Use only 3.3 V UART logic.

Testpoints and Optional Pads

Use TP2, TP7, or TP9 as ground reference when measuring nearby signals. Firmware defaults leave reservoir and flow sensing disabled.

TP	Net / purpose	Where on board	Expected / how to use it
TP1	+5V rail test pad.	top-right; x=107.03 mm, y=57.98 mm	+5 V when USB/5 V supply is present.
TP2	GND reference test pad.	top-right; x=107.03 mm, y=61.53 mm	0 V reference ground.
TP3	+3V3 logic rail test pad.	top-right; x=107.03 mm, y=65.08 mm	+3.3 V after AP63203 buck is working.
TP4	PUMP_GATE test pad.	top-right; x=107.03 mm, y=68.63 mm	Normally LOW at reset/boot; about 3.3 V only when pump command is active.
TP5	MOISTURE_ADC test pad.	middle-right; x=107.03 mm, y=72.18 mm	Analog moisture reading, 0 to 3.3 V max.
TP6	Optional reservoir switch signal pad; firmware default disabled.	bottom-right; x=98.58 mm, y=109.63 mm	Optional switch signal; logic depends on future reservoir switch firmware setting.
TP7	Optional ground pad for reservoir switch wiring.	bottom-right; x=101.08 mm, y=109.63 mm	0 V reference ground.
TP8	Optional +3V3 pad for optional sensor wiring.	bottom-center; x=86.63 mm, y=109.63 mm	+3.3 V optional sensor power pad.
TP9	Optional ground pad for flow/optional sensor wiring.	bottom-center; x=89.13 mm, y=109.63 mm	0 V reference ground.
TP10	Optional flow pulse signal pad; must be 3.3 V safe if used.	bottom-center; x=91.63 mm, y=109.63 mm	Optional flow pulse input; must never exceed 3.3 V at ESP32 pin.

Bring-Up Measurement Quick Checks

Check	Meter setup	Pass condition
No power shorts	Resistance/continuity with board unpowered	No direct short from +5V to GND or +3V3 to GND.
USB input rail	TP1 to TP2	About +5 V.
Logic rail	TP3 to TP2	About +3.3 V. Stop if missing before installing/testing ESP32.
Pump gate reset state	TP4 to TP2 during reset/boot	LOW/near 0 V. Pump must not turn on during reset.
Moisture input	TP5 to TP2 with sensor connected	Analog voltage between 0 and 3.3 V.